TRACE: Trajectory-based Activation Change Estimation for Task-specific Data Selection

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Appendix

Evaluation Tasks Details

Table 1 summarizes the statistics of the three benchmarks used for our evaluation. Our suite is designed to be diverse, covering a wide range of capabilities. BBH comprises 27 distinct, challenging reasoning tasks, while MMLU spans 57 domains of knowledge-based question answering. For multilingual capabilities, we use TyDiQA, which includes QA tasks in 9 different languages. Performance on BBH and MMLU is measured using the standard lm-evaluation-harness, while for TyDiQA, we adopt the evaluation pipeline from LESS for consistency. Notably, the TyDiQA benchmark provides merely nine examples in its validation set, presenting a particularly challenging low-resource scenario for data selection.

Task	$\lvert\mathcal{D}_{val}\rvert$	$\lvert\mathcal{D}_{test}\rvert$	Shot	Metric
BBH	81	920	3	Exact Match
MMLU	285	18721	5	Accuracy
TYDiQA	9	5077	1	F1

Table 1: Details of all evaluation tasks.

Training Details

Warmup Data Construction We utilize an A100 cluster to conduct experiments. For the TRACE method, we investigate two distinct strategies for constructing the warmup dataset, $\mathcal{D}_{warmup}$:

Random Sampling: We create three separate warmup sets by randomly sampling 5% of the source corpus using different random seeds (3, 6, and 42). To ensure a fair and direct comparison, these same three subsets are also used as the training data for our Random baseline method.

K-Means Clustering-based Sampling: To create a more structured and diverse warmup set, we first generate data representations using the RDS method. We then apply k-means clustering with 100 clusters (seed=42). The final warmup set is constructed by selecting the 100 instances closest to each of the 100 cluster centroids, resulting in a total of 10,000 diverse examples.

Layer Selection Protocol: Our experimental procedure involves two stages. First, we conduct a preliminary analysis using the k-means-derived warmup data. We perform the full TRACE selection process and evaluate its performance across all layers of the model. Based on these results, we identify a contiguous region of best-performing layers to serve as our "target layers." Subsequently, for our main reported results, we use the models warmed up with the randomly sampled data and apply the TRACE computation and data selection exclusively on these pre-identified target layers.

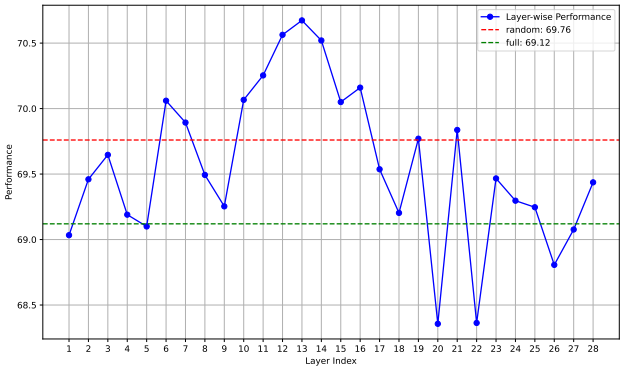


Figure 1: Layer Effectiveness on Qwen-2.5-7B.

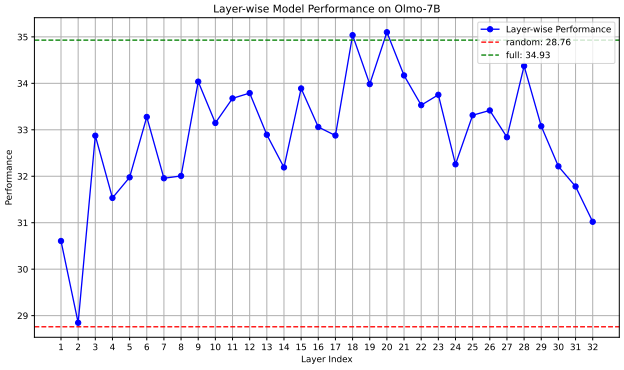


Figure 2: Layer Effectiveness on OLMO-7B.

Layer effectiveness

Figure 1 and 2 illustrate the layer-wise performance of TRACE on the OLMo-7B and Qwen-2.5-7B models, respectively. The results reveal a consistent trend that corroborates our findings on LLaMA-3.1-8B: while the specific optimal layer varies across architectures, the intermediate layers (approx. 10-15) consistently emerge as the most effective region for TRACE. This finding aligns with a growing body of recent work on the functional specialization of transformer layers .

More Experiment Result

model	Base	Random	Full	RDS	LESS	TRACE
LLama-3.1-8B	53.68	55.35	63.68	61.49	57.47	64.37
Qwen-2.5-7B	83.62	75.74	75.06	77.79	76.27	76.42
Olmo-7B	7.28	8.97	22.74	9.10	1.97	22.37

Table 2: Performances of different methods on GSM8K

model	layer 1	layer 2	layer 3	layer 4	layer 5
LLama-3.1-8B	94.42	90.74	88.63	91.96	86.82
Qwen-2.5-7B	81.90	64.19	86.02	84.00	81.16
Olmo-7B	95.83	95.66	96.14	96.50	94.97

Table 3: Overlap rate (%) between TRACE and Neuron Activation similarity on GSM8K.

Experimental result on GSM8K We also conduct experiment on GSM8K for further analysis. We randomly select 128 data from GSM8K training set as validation data. As shown in Table 2, our TRACE outperforms all the baseline. Moreover, on the homogeneous GSM8K benchmark, our TRACE method's selections notably converged with those from Neuron activation similarity(NAS) (see Table 3 and Table 4, NAS is a simple method that we utilize neuron activation to compute similarity and select data), both successfully identifying relevant math problems. This success contrasts with the failure of gradient-based methods, which selected irrelevant data. This reveals two insights: 1) for uniform tasks, the core signal is strong enough to be captured by representation-based methods, and 2) the loss gradient is a particularly noisy and brittle proxy for complex reasoning, making gradient-based selection unreliable for such tasks.

This phenomenon can be explained by the fundamental difference between the spaces in which these methods operate. Gradient-based methods perform their similarity comparison in the parameter space. A gradient is a high-dimensional vector dictating how to mechanically adjust every model weight to reduce loss. This space is vast and lacks a clear semantic structure; for complex generative tasks, two logically similar problems might require vastly different parameter updates, leading to low gradient similarity. In contrast, our TRACE operates within the representation space. This space is inherently semantic, as it is where the model encodes meaning. By measuring the shift within this structured, semantically-grounded space, TRACE captures a signal that is more robustly aligned with the task's core logic, rather than the brittle mechanics of parameter updates.

	BBH	**MMLU**	**TyDiQA**	**AVG**
Qwen-2.5-7B				
Random*	62.79	73.01	**73.47**	69.76
Full*	62.11	72.95	72.3	69.12
llama-data	**64.71**	**73.02**	72.39	**70.04**
Olmo-7B				
Random*	29.93	27.18	29.17	28.76
Full*	30.38	**42.46**	31.96	**34.93**
llama-data	19.06	40.72	**34.82**	31.53

Table 4: Performance of transferability. We use the data selected by LLama-3.1-8B to finetune Olmo-7B and Qwen-2.5-7B.

Moreover, on the GSM8K benchmark, we observed that RDS selects a mixed-quality dataset, comprising both relevant math problems and a significant number of irrelevant descriptive texts (see Table 6). Our analysis indicates this is because RDS incorrectly prioritizes the semantic similarity of the problem's narrative scenario over its mathematical reasoning structure. By conflating these two signals into a single representation, RDS is easily misled into selecting other narrative-style examples that lack the essential procedural component, highlighting a key failure mode for static, representation-based methods on compositional tasks.

Transferability To assess the transferability of our method, we conducted a cross-model evaluation. We took the optimal data subset selected by TRACE using the LLaMA-3.1-8B model and used it to fine-tune the other two models in our testbed. As shown in Table 4, the results show that, in most cases, this transferred subset significantly outperforms the random selection baseline for the target models. This indicates that the data selected by TRACE exhibits strong generalizability across different model architectures. We hypothesize that this is because TRACE captures not only a model-specific optimization trajectory but also fundamental semantic representations. While the specifics of these representations differ, the underlying conceptual geometry is often similar across well-trained foundation models, allowing the data selected by TRACE to be effectively transferred.

Validation Example of GSM8K	
User: Archer caught eight fish from the lake to sell in the market. When he reached the market, he sold the fish faster than he had anticipated and decided to go back to the lake and catch more fish. He caught 12 more fish in the second round than he had caught earlier. The demand was even greater, and he had to close the day by catching 60% more fish than the number he had caught in the second round and sold all of them in the market. How many fish did he catch that day? **Assistant:** He caught 12 more fish in the second round than he had caught earlier, a total of 12+8=12+8=20 fish. The total number of fish he caught in the two rounds is 20+8=20+8=28 In the third round, Archer caught 60/100*20=60/100*20=12 fish which were 60% more fish than the number he had caught in the second round. This means he caught a total of 12+20 = 12+20=32 fish Archer caught a total of 28+32=28+32=60 fish on that day. 60	
RDS(All data)	NAS(All data)
User: 2 pounds of steak will cost Ken 2 * 7 = 14 dollars. Subtracting that amount from $20 will give 20 - 14 = 6 dollars. Ken will therefore get 6 dollars back.is the reasoning for what question and answer pair? **Assistant:** Q: At Ken's local store, a pound of steak is going for $7. He decides to get two pounds. At the till, he pays using a $20 bill. How much money will he get back? A: 6	**User:** [Examples...] Mark is a lawyer who works at the county courthouse every day. It takes him 5 minutes to find parking and 3 minutes to walk into the courthouse. 2 days of the week it takes him 30 minutes to get through the metal detector and the other 3 days are less crowded so he gets through in 10 minutes. How long does he spend on all these activities in a week with 5 work days? **Assistant:** First find how long Mark spends on parking and walking each day: 5 minutes + 3 minutes = 8 minutes. Then multiply that number by his number of workdays to find the total walking and parking time: 8 minutes / day * 5 days / week = 40 minutes / week. Then multiply the number of long - wait days by the time of each long wait: 2 days / week * 30 minutes / day = 60 minutes / week. Then multiply the number of short - wait days by the time of each short wait: 3 days / week * 10 minutes / day = 30 minutes / week. Then add the length of time Mark spends on each activity to find the total time spent: 30 minutes / week + 60 minutes / week + 40 minutes / week = 130 minutes / week. The answer is 130.
LESS(All data)	TRACE(All data)
User: Hi! Having 2023 in mind, I need you to continue the series (list all Mondays): 6/01/2023 13/01/2023 20/01/2023 ... **Assistant:** 27/01/2023 03/02/2023 10/02/2023 17/02/2023	**User:** Amanda charges $20.00 per hour to help clean out and organize a person's home. She has 5 1.5 hours appointments on Monday, a 3-hours appointment on Tuesday and 2 2-hours appointments on Thursday. On Saturday, she will spend 6 hours at one client's house. How much money will she make this week? Let's be accurate as possible. **Assistant:** She has 5 1.5 hour appointments on Monday for a total of 5 * 1.5 = 7.5 hours. She has 2 2 hour appointments on Thursday for a total of 2 * 2 = 4 hours. She is working 7.5 hours on Monday, 3 hours on Tuesday, 4 hours on Thursday and 6 hours on Saturday for a total of 7.5 + 3 + 4 + 6 = 20.5 hours. She charges $20.00 an hour and is working 20.5 hours a week for a total of 20 * 20.5 = $410.00. The answer: 410.

Table 5: Selected Data Comparison. "all data": data was selected across all validation data. The data are selected by top 30 indices.

Validation Example of GSM8K	
User: Archer caught eight fish from the lake to sell in the market. When he reached the market, he sold the fish faster than he had anticipated and decided to go back to the lake and catch more fish. He caught 12 more fish in the second round than he had caught earlier. The demand was even greater, and he had to close the day by catching 60% more fish than the number he had caught in the second round and sold all of them in the market. How many fish did he catch that day? **Assistant:** He caught 12 more fish in the second round than he had caught earlier, a total of 12+8=12+8=20 fish. The total number of fish he caught in the two rounds is 20+8=20+8=28 In the third round, Archer caught 60/100*20=60/100*20=12 fish which were 60% more fish than the number he had caught in the second round. This means he caught a total of 12+20 = 12+20=32 fish Archer caught a total of 28+32=28+32=60 fish on that day. 60	
RDS(All data)	RDS(Single data)
User: A boy riding a bike does not have to be practicing.is the reasoning for what question and answer pair? **Assistant:** Q: Premise: "A boy with a blue helmet is riding a bike." Hypothesis: "A boy practicing how to ride a bike." Is the hypothesis entailed by the premise? Options: - yes - it is not possible to tell - no A: it is not possible to tell	**User:** On a scale of 1-5 (with 1 being least favorable and 5 being most favorable), how would you rate this review? "Love it Very good" Answer: **Assistant:** 5

Table 6: Selected Data Comparison. "All data": data was selected across all validation data. "Single data": data was select by the example validation data. The data is selected by top 30 indices.